Reproducing MelBERT for Metaphor Detection and Testing Its Generalisation to New Domains

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Abstract

We reproduce MelBERT (Choi et al., 2021), a RoBERTa-based word-level metaphor detector that builds two linguistic theories of metaphor identification into its architecture, and test how well it generalises beyond its benchmark. With the authors' official code and a three-line compatibility patch, the released checkpoint scores 78.8 F1 on VUA-18, and our retrained model 78.2, against the reported 78.5. Results on VUA-Verb, MOH-X and TroFi are also within 1.6 F1 of the paper. We then build a new evaluation set from two domains absent from the benchmarks, Amazon product reviews and arXiv abstracts: 150 sampled sentences (1,436 content-word targets), of which a stratified random subset of 40 sentences (388 targets) was hand-annotated with MIPVU. An LLM labelled the same targets independently as a second rater. On the new data MelBERT drops to 36.8 F1 on reviews and 31.6 on arXiv, against 72.1 on the matching parts of speech of VUA-18. It over-predicts metaphor two-fold, mostly on delexical verbs and abstract academic vocabulary that VUA's conventions treat as metaphorical. Human–LLM agreement was low ($\kappa = 0.15$), which shows how hard consistent metaphor annotation is outside VUA. Code and data: https://github.com/shravaneekale/melbert-reproduction.

1 Introduction

Metaphor is pervasive: roughly one word in eight in the VU Amsterdam Metaphor Corpus is metaphorical (Steen et al., 2010). Systems that read figurative language literally make errors in sentiment analysis, translation and dialogue. MelBERT (Choi et al., 2021) is a widely used neural metaphor detector. This project had two goals: **(G1)** replicate MelBERT's published results with the authors' code and data, and **(G2)** test whether MelBERT generalises to genres its training data does not contain, using newly annotated data.

2 The Paper and Why We Chose It

Task and model Given a sentence and a target word, the task is to decide whether the word is used metaphorically. MelBERT (Choi et al., 2021)[1] encodes the sentence with RoBERTa-base (Liu et al., 2019) and operationalises two theories from linguistics:

- **MIP** (Pragglejaz Group, 2007; Steen et al., 2010): a word is metaphorical if its contextual meaning contrasts with its basic meaning. MelBERT compares the contextualised embedding of the target with an embedding of the target word encoded *alone*, which approximates its basic meaning.
- **SPV** (Wilks, 1978): a word is metaphorical if it violates the selectional preferences of its context. MelBERT compares the sentence embedding with the contextualised target embedding.

The two signals are computed by separate layers over the same encoder and combined only before the final classifier (*late interaction*). The input also marks the target's POS tag and its local context (the clause between commas) using extra segment embeddings. Ablations in the paper show that MelBERT outperforms its MIP-only and SPV-only variants, so the two theories provide complementary signals.

Evaluation framework Each target word is classified as metaphorical (M) or literal (L), and the paper reports precision, recall and F1 on the M class. Accuracy is not used because only about 12% of VUA tokens are metaphorical, so predicting "literal" everywhere already gives about 88% accuracy. MelBERT reports 78.5 F1 on VUA-18 (P 80.1, R 76.9), 75.7 on VUA-Verb and 72.3 on

[1] https://aclanthology.org/2021.naacl-main.141

VUA-20. In zero-shot transfer from VUA-20 it scores 79.2 F1 on MOH-X and 62.0 on TroFi, outperforming RoBERTa baselines and earlier models. All numbers are means over five seeds.

Justification (i) *Venue*: NAACL is ranked **A** in the CORE/ICORE conference rankings, a top-tier venue for computational linguistics. (ii) *Impact*: the paper has about 100 citations (Semantic Scholar, Nov. 2026), and later metaphor-detection work still uses it as a standard baseline. (iii) *Code*: the authors' official implementation is public[2]. The first author's GitHub account hosts it, along with pre-processed data and a trained checkpoint. (iv) *Data*: all four benchmarks are public, and the authors also release them pre-processed into the code's input format, which removes a major source of replication error. (v) *Feasibility*: our proposal pilot ran the full pipeline end to end on a laptop after a one-line fix (§4). Together these make MelBERT a paper whose results *should* be reproducible, which makes testing that claim worthwhile.

3 Original Datasets

The paper's main benchmark is the **VU Amsterdam Metaphor Corpus** (VUAMC; Steen et al., 2010). It contains about 190k words of British National Corpus text from four genres (academic, news, fiction and conversation), annotated word by word with MIPVU. We use the two shared-task releases: **VUA-18** (Leong et al., 2018), with the split of Mao et al. (2019),[3] and **VUA-20** (Leong et al., 2020), which adds more tokens and parts of speech.[4] The paper also breaks the VUA-18 test set down by genre and by POS (verb, noun, adjective, adverb) to analyse where the model succeeds. **VUA-Verb** is the verb subset of the VUA-18 test set. Two small, verb-only sets are used for test-only transfer: **MOH-X** (Mohammad et al., 2016), built from WordNet example sentences, and **TroFi** (Birke and Sarkar, 2006), built from Wall Street Journal sentences (both available from the same repository as VUA-18). Both follow annotation conventions different from VUA's, which is why the paper uses them to test generalisation (Table 1).

Format The authors distribute pre-processed TSV files[5] with one row per (sentence, target word).

[2] https://github.com/jin530/MelBERT

[3] https://github.com/RuiMao1988/Sequential-Metaphor-Identification

[4] https://github.com/YU-NLPLab/DeepMet

[5] Linked from the repository README (Google Drive).

Dataset	#targets	%M	#sent.
VUA-18 train	116,622	11.2	6,323
VUA-18 dev	38,628	11.6	1,550
VUA-18 test	50,175	12.4	2,694
VUA-20 train	160,154	12.0	12,109
VUA-20 test	22,196	17.9	3,698
VUA-Verb test	5,873	30.0	2,694
MOH-X	647	48.7	647
TroFi	3,737	43.5	3,737

Table 1: Benchmark statistics (from the MelBERT repository). %M = percentage of metaphorical targets.

```
index             label sentence                  POS  w_index
a3m-fragment02 45 0     Design: Crossed lines ... NOUN 0
a3m-fragment02 45 1     Design: Crossed lines ... ADJ  1
a3m-fragment02 45 1     Design: Crossed lines ... NOUN 2
```

Figure 1: VUA-18 rows in MelBERT's input format (sentence truncated). Full sentence: *Design: Crossed lines over the toytown tram: City transport could soon be back on the right track, says Jonathan Glancey*; *Crossed* and *lines* are metaphorical.

The columns are a sentence id, the label (1 = metaphorical), the whitespace-tokenised sentence, the target's Universal POS tag, and the target's index in the whitespace-split sentence (Figure 1). VUA-18/20 include *every* token as a target, function words too. MOH-X and TroFi use the same columns but have exactly one target verb per sentence, e.g. *The invaders **spread** their language all over the country* (label 1). They are distributed as ten cross-validation folds.

4 Replication of the Original Work

Code and environment We cloned the official repository at commit `11b0f78` and changed only three lines (`setup/patch_melbert.py`). (1) `AdamW` is imported from `torch.optim`, because `transformers` no longer provides it. (2) The device selection falls back to Apple-Silicon MPS. (3) `torch.load` is given a `map_location`, so GPU-trained checkpoints load on a laptop. We pinned `transformers` 4.46.3, because the 2021 code depends on 4.x tokenizer behaviour. The original pins (Python 3.7, PyTorch 1.6, `transformers` 4.2.2) no longer install on current hardware. The released checkpoint was scored on a Google Colab NVIDIA T4 GPU. Full retraining ran on an Apple-Silicon MacBook Pro (PyTorch 2.15, MPS backend) and took 5 h 44 min (about 1.9 s per batch of 32), plus 12 minutes for each test-set evaluation. All scripts are in our repos-

itory.[6]

Procedure We ran three experiments: **(R1)** scoring the authors' *released checkpoint* on the test sets, which checks code and data without retraining; **(R2)** *retraining* MelBERT from RoBERTa-base on the full VUA-18 training set with the paper's settings (3 epochs, batch 32, learning rate 3×10^{-5} with linear warm-up over 2 epochs, dropout 0.2, maximum length 150, seed 42); **(R3)** our proposal's reduced-budget runs on an Apple-Silicon laptop. MOH-X and TroFi are scored zero-shot on the union of their ten test folds. A per-token prediction script (`src/predict.py`) reuses the official model and feature code and keeps every prediction for later analysis.

Results Table 2 compares our numbers with the paper's. The authors' released checkpoint scores 78.8 F1 on VUA-18 (P 79.4, R 78.3), 75.6 on VUA-Verb, 79.2 on MOH-X and 61.0 on TroFi, within 1 F1 of every reported number. The released code, data and model therefore reproduce the published results. Our retrained model scores 78.2 on VUA-18 (P 79.7, R 76.7, accuracy 93.9), against the reported 78.5 (P 80.1, R 76.9). Test-set F1 after epochs 1, 2 and 3 was 35.8, 71.8 and 78.2. The slow start follows from the schedule: the learning rate warms up linearly for two epochs and only peaks at the end of epoch 2, so most of the learning happens in the last epoch. On VUA-Verb the retrained model scores 75.3 (reported 75.7). Zero-shot it scores 80.8 on MOH-X and 62.3 on TroFi (reported 79.2 / 62.0), despite being trained on the smaller VUA-18 rather than VUA-20. Scoring the released checkpoint on the Mac (MPS) gave exactly the same numbers as on the Colab GPU, so inference does not depend on the hardware. TroFi stays far below the other sets for every model (precision about 52): its literal/non-literal labels follow different guidelines from MIPVU, and a VUA-trained model over-predicts metaphor there. Table 3 shows that training budget matters a great deal. With 3k sentences and the default two-epoch warm-up, the model first collapses to the majority class (0 F1 at 86% accuracy), exactly the failure mode that makes accuracy uninformative. With 10k sentences it reaches 68.2 F1.

How close Our retrained F1 is 0.3 points below the paper's (0.4% relative), and the released checkpoint is 0.3 above it. The paper does not report seed variance, but bagging over seeds changes its F1 by 1.3 points (78.5 to 79.8), so differences of about one point are within normal run-to-run variation. Precision and recall are each within 0.4 points of the paper, so the small difference is not a shift in the precision–recall balance. Overall, the main VUA-18 result **reproduces**, both from the released checkpoint and from retraining with the released code.

Why the numbers can differ Reading the code against the paper turned up several differences that matter for reproducibility. (1) *Seeds*: the paper reports a mean over five seeds; we trained one seed (42), which our compute budget allowed. (2) *Model selection*: the paper says hyperparameters were tuned on the VUA-18 development set, but the released code evaluates on the *test* set after every epoch and never loads `dev.tsv`. For VUA it keeps the last epoch, so there is no test-set selection. For MOH-X/TroFi cross-validation, though, it keeps the epoch with the best *test-fold* F1, which slightly inflates those numbers. (3) *Class weighting*: the code weights the metaphor class $3\times$ in the loss (`class_weight = 3`), which the paper does not mention. This trades precision for recall. (4) *Software*: newer PyTorch/`transformers` versions change numerics and tokenisation slightly, and GPU training is not bit-wise deterministic. (5) *Transfer*: our MOH-X/TroFi model was trained on VUA-18, not VUA-20. (6) *Test-set size*: the VUA-18 test file contains 43,947 scored targets, fewer than the 50,175 tokens listed in the repository's statistics table, so published token counts and scored targets differ.

5 Construction of New Data

Choice of domains The proposal listed five candidate corpora. We kept the two that are freely licensed, need no scraping, and differ from VUA along clear axes. **(a) Amazon product reviews** (`fancyzhx/amazon_polarity` on Hugging Face; Zhang et al., 2015, derived from McAuley and Leskovec, 2013) are informal, evaluative, user-written text, a register VUA lacks. We expect personification (*my laptop **crawls***) and hyperbole there. **(b) arXiv abstracts** (`gfissore/arxiv-abstracts-2021`, CC0; Clement et al., 2019) are technical prose. They are more terminology-dense than VUA's academic genre, so we expect few but subtle,

[6] https://github.com/shravaneekale/melbert-reproduction

Test set	Paper P	Paper R	Paper F1	Ours P	Ours R	Ours F1	Official ckpt P	Official ckpt R	Official ckpt F1	ΔF1
VUA-18	80.1	76.9	78.5	79.7	76.7	78.2	79.4	78.3	78.8	-0.3
VUA-Verb	78.7	72.9	75.7	75.1	75.6	75.3	73.2	78.0	75.6	-0.4
MOH-X†	79.3	79.7	79.2	79.8	81.9	80.8	80.3	78.0	79.1	+1.6
TroFi†	53.4	74.1	62.0	52.3	77.2	62.3	52.2	73.5	61.0	+0.3

Table 2: Replication (MelBERT, RoBERTa-base). Paper = Choi et al. (2021), mean of 5 seeds. Ours = retrained with the official code, seed 42 (R2). Official ckpt = the authors' released VUA-18 checkpoint (R1). †Zero-shot: the paper trains on VUA-20 for these sets; we use our VUA-18 model.

Training data (laptop)	Ep.	P	R	F1
3k sentences, warm-up 2	1	0.0	0.0	0.0
	3	31.2	30.2	30.7
10k sentences, warm-up 0.5	1	31.2	19.3	23.9
	4	69.6	66.7	68.2

Table 3: Reduced-budget runs (R3, from our proposal; Apple M-series, MPS), evaluated on the full VUA-18 test set.

```
index    label sentence                          POS  w_index
rev0014  0     As others have written, the ... NOUN 5
rev0014  1     As others have written, the ... VERB 7
arx0017  1     Here we test these constraints ... NOUN 8
```

Figure 2: New data in MelBERT's input format (`data/NEW/all/test.tsv`). Sentences: *As others have written, the package it **comes** in **sucks*** (*package* literal, *comes* and *sucks* metaphorical); *Here we test these constraints numerically in the **framework** of the Abelian-Higgs model…* (*framework* metaphorical).

highly conventional metaphors (*a **framework**, **tackle** a problem*). These two domains test both directions of domain shift: whether MelBERT over-predicts metaphor in literal text, and whether it misses informal figurative language. We dropped lyrics (copyright), live-forum scraping (platform terms and API restrictions) and sports commentary (cleaning cost).

From unlabelled corpora to labelled data Neither corpus has metaphor labels, so every label in our new data comes from our own MIPVU annotation, not from the source datasets. The 60 *synthetic* sentences shown in our progress video were discarded, and every sentence used here is real, sampled text.

Sampling and processing We streamed each corpus in a seeded random order. To maximise diversity we took at most one sentence per document. For arXiv we balanced five subject areas (physics, mathematics, computer science, statistics, quantitative biology). Sentences were kept if they had 8–40 tokens, were mostly alphabetic, contained no URLs, e-mail addresses, handles or LaTeX maths, and had no PERSON named entity (a privacy filter, using spaCy; Honnibal et al., 2020). Exact duplicates were removed. We then whitespace-tokenised each sentence exactly as MelBERT does and assigned Universal POS tags (as in VUA-18) by aligning spaCy tokens to whitespace tokens. Every *content word* (verb, noun, adjective, adverb; no proper nouns or auxiliaries) became a target. Content words carry most of the metaphors that matter for downstream use, and fully annotating function words was beyond our budget. For a like-for-like comparison we therefore also report VUA-18 restricted to the same four POS. Figure 2 shows example items (code: `src/sample_corpus.py`, `src/build_targets.py`).

Annotation Labels follow MIPVU (Steen et al., 2010), the procedure behind VUA itself, so our gold standard uses the same definition of metaphor as MelBERT's training data. For each target the annotator (1) determines its contextual meaning, (2) identifies its basic meaning (more concrete, bodily or precise; checked in Longman/Macmillan as MIPVU prescribes), and (3) labels it M if the two contrast but are related by comparison (guidelines: `annotation/guidelines.md`). A purpose-built terminal tool showed one target at a time and recorded labels, a low-confidence flag and notes.

Scope. Within the project's time budget, the author hand-annotated a **stratified random subset of 40 sentences** (20 per domain, seed 8240), giving 388 targets. All 150 sentences and 1,436 targets are released, with LLM labels for every target as a silver standard.

Second rater. No second human annotator was available, so an LLM (LLM1: Claude Opus 5.5 through claude.ai, 5 November 2026) labelled every target independently. It received the MIPVU instructions in `src/llm_judge.py` and never saw the human labels. The human annotation was

Domain	Sent.	Targets	%M	V	N	Adj	Adv
Reviews	20	141	8.5	17.5	8.2	3.4	0.0
arXiv	20	247	8.5	10.4	10.4	3.7	0.0
All	40	388	8.5	13.6	9.8	3.6	0.0

Table 4: Hand-annotated gold standard: sentences, targets, and % metaphorical overall and by POS.

Subset	n	Agree.	κ	%M (H/LLM)
All	388	0.88	0.15	6.7/8.0
Reviews	141	0.87	0.10	7.8/8.5
arXiv	247	0.89	0.18	6.1/7.7
Verb	88	0.86	0.20	4.5/13.6
Noun	183	0.86	0.17	8.7/9.3
Adj	83	0.90	−0.04	7.2/2.4
Adv	34	1.00	–	0.0/0.0

Table 5: Agreement between the human annotator (H) and the LLM rater before adjudication.

completed without seeing the LLM's labels.

Pilot. A first, pilot pass by the author labelled only 0.8% of targets as metaphorical, because conventionalised metaphors (*the battery **died***) were treated as literal. That pass was discarded and the guidelines re-read before the final annotation. The pilot labels are kept in the repository.

Adjudication. Every target where the author and the LLM disagreed (45 of 388) was resolved by the author. For each one the LLM was asked for a MIPVU analysis (basic meaning, contextual meaning, reason), which the author reviewed before recording the final label and reason (`data/annotation/adjudication.csv`).

Agreement and statistics Table 5 reports Cohen's κ (Cohen, 1960) between the author and the LLM *before* adjudication. Raw agreement is high (88.4%) because most targets are literal, but chance-corrected agreement is only $\kappa = 0.15$ (slight; Landis and Koch, 1977). Both raters found a similar *rate* of metaphor (6.7% vs. 8.0%) but mostly picked *different* words: they agreed on only 6 metaphors, against 20 marked only by the author and 25 only by the LLM. The author's extra labels were mainly unusual or technical words used literally (*oversize*, *gameplay*, drill *bits*). The LLM's were conventional metaphors (*framework*, *regime*, *agreement*, Nirvana *released* an album, the package *sucks*). Adjudication sided with the LLM in 43 of the 45 cases, so the gold standard is closer to the LLM's reading of MIPVU than to the author's original one. We return to this in the limitations. For comparison, the VUAMC reliability tests with four trained annotators reached Fleiss' $\kappa \approx 0.85$ (Steen et al., 2010); the gap shows how much training MIPVU needs.

Table 4 summarises the gold standard: 8.5% of targets are metaphorical in both domains, against 15.9% for VUA-18 test content words. Verbs are the most metaphorical part of speech, as in VUA. No adverb was labelled metaphorical.

6 Results on New Data

We ran our retrained VUA-18 model (R2) and the released checkpoint (R1) unchanged on the 388 gold targets. Confidence intervals come from a sentence-level (cluster) bootstrap with 1,000 resamples (Efron and Tibshirani, 1993), because targets in the same sentence are not independent.

Overall Table 6 gives the main result. On VUA-18 content words our model scores 72.1 F1. On the new domains it scores 36.8 (reviews) and 31.6 (arXiv), a drop of 35–40 points. The test sets are small, so the bootstrap intervals are wide, but even their upper bounds (55.2 and 43.9) lie far below the VUA interval [70.8, 73.3], so the drop is not an artefact of sample size. The released checkpoint behaves almost identically (31.4 vs. 33.3 F1 overall), so the drop is a property of MelBERT, not of our training run. Recall stays moderate (about 58%), but precision collapses to 22–27%. MelBERT labels 20.9% of targets as metaphorical where the gold standard has 8.5%: it over-predicts about 2.5-fold.

Reviews 19 of the 26 positive predictions are false positives. Most are *delexical or conventional* uses: *make* (up for), *made* (easier), *have* (no talent), *great* (action sequences), a *couple* of, a *little* bit, *that* much, it *teaches*. In VUA's MIPVU conventions many such words count as metaphorical (e.g. *great* meaning "very good" contrasts with "large"), so MelBERT has learned a broader notion of metaphor than our annotation applied. Other false positives are literal hobby vocabulary unlike anything in VUA (fishing reel *cranked*, *baitfeeding*, *riffs*). The five misses are informal or idiomatic metaphors: *burning* a CD, *on hand*, the *best of places*, *old school*, the package it *comes* in. MelBERT gave these a metaphor probability below 0.3.

Test set	n	%M gold	%M pred	P	R	F1	95% CI	F1 (R1)
VUA-18 test (V/N/Adj/Adv)	24,849	15.9	14.9	74.5	69.7	72.1	[70.8, 73.3]	–
Reviews (new)	141	8.5	18.4	26.9	58.3	36.8	[18.8, 55.2]	33.3
arXiv (new)	247	8.5	22.3	21.8	57.1	31.6	[17.1, 43.9]	30.6
New (both)	388	8.5	20.9	23.5	57.6	33.3	[22.4, 43.6]	31.4

Table 6: MelBERT on the new domains (our retrained model R2; the last column is the released checkpoint R1). The first row restricts the VUA-18 test set to the same four POS as the new data. %M pred > %M gold means the model over-predicts metaphor.

POS	Paper (VUA-18)	VUA-18	Reviews	arXiv
Verb	75.1	75.6 (9247)	50.0 (40)	34.5 (48)
Noun	70.7	68.7 (8583)	18.2 (49)	29.3 (134)
Adj	64.4	65.4 (3965)	33.3 (29)	33.3 (54)
Adv	74.6	74.6 (3054)	0.0 (23)	0.0 (11)

Table 7: F1 by POS (number of targets in brackets). Paper = Table 5 of Choi et al. (2021). No adverb in the new data is gold-metaphorical, so its F1 is 0.

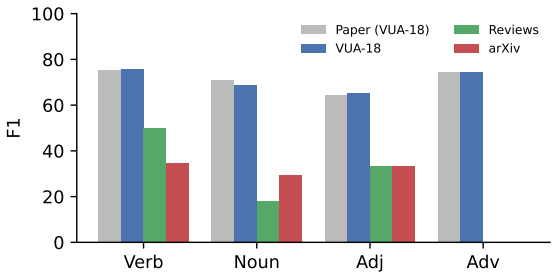


Figure 3: F1 by POS: paper (VUA-18), our replication (VUA-18 test) and the two new domains.

arXiv Errors are dominated by 43 false positives on academic and technical vocabulary: *show*, *obtain*, *introduce*, *implies*, *aim*, *application*, *space*, *direct*, *positive*, *larger*, *force*, *model*, plus physics terms (*vortices*, *array*, *advected*). Many are exactly the abstract uses of concrete words that VUA's academic genre marks as metaphorical (the paper reports 83.9 F1 on VUA academic), so this is partly a difference in annotation convention. Terminology-dense abstracts also contain many words whose technical sense is now basic, and MelBERT over-applies the MIP signal to them. Conversely, it missed conventional abstract-noun metaphors that our annotation counts (*framework*, *regime*, *agreement*, *census*, software *agent*, computing *environments*, *hierarchy*), often with near-zero probability.

By POS Table 7 and Figure 3 show that on VUA our replication matches the paper's per-POS results to within 2 F1. Out of domain, verbs hold up best in reviews (50.0) and nouns degrade most (18.2 in reviews, 29.3 in arXiv). In VUA the hardest POS is adjectives (paper: 64.4); on the new data it is nouns, consistent with abstract nouns in technical prose being where the two annotation conventions differ most.

LLM-as-judge We used LLMs as judges in three ways. **(1) Independent rater**: LLM1 judged the metaphoricity of every target. Its agreement with the author on the hand-annotated subset is $\kappa = 0.15$, so an LLM cannot simply replace a human annotator for MIPVU, and a single non-expert annotator is not a reliable reference either. **(2) Adjudication aid**: for the 45 disagreements LLM1 supplied MIPVU analyses that the author reviewed. **(3) Reference-free evaluation**: could LLM judgements replace human labels when scoring MelBERT on a new domain? A second LLM from a different provider (LLM2: ChatGPT, OpenAI, accessed through chatgpt.com on 5 November 2026) had also labelled every target blind, with the same MIPVU instructions. We scored MelBERT's predictions against each LLM's labels instead of the human gold (`src/llm_reference_eval.py`, Table 8). The two judges give very different answers. Judged by LLM1, MelBERT scores 32.1 F1, close to the human-gold 33.3. But this is circular: adjudication mostly followed LLM1, and on reviews LLM1 and the gold agree perfectly. Judged by the independent LLM2, which marks only 1.3% of targets as metaphorical, MelBERT scores just 7.0 F1, because almost every positive prediction becomes a false positive. LLM2 agrees with the human gold at $\kappa = 0.14$. LLM-as-judge estimates of absolute performance therefore depend more on which LLM judges, and on how liberally it applies MIPVU, than on the system being judged. We would use LLM judges only to pre-screen items for human review, not to report scores.

How the new results relate to the replication In domain, the replication holds to within 0.3 F1 on

Reference	LLM1		LLM2		Human
Test set	%M	F1	%M	F1	F1
Reviews	8.5	36.8	2.1	6.9	36.8
arXiv	7.7	29.7	0.8	7.0	31.6
Both	8.0	32.1	1.3	7.0	33.3

Table 8: MelBERT's F1 on the new data when each LLM's blind labels serve as the reference, against the human gold. %M = each judge's metaphor rate. The LLM1 estimate is circular, because adjudication drew on LLM1.

VUA-18 and within 1.6 F1 on every benchmark. Out of domain, the same model loses 35–40 F1, mostly through over-prediction. The new-data evaluation differs from the replication in four ways: (i) only content words are targets; (ii) the labels come from one non-expert annotator assisted by an LLM, not from trained MIPVU annotators, and are more conservative about conventional metaphor than VUA (8.5% vs. 15.9% metaphorical); (iii) agreement was low, which caps the F1 any system can attain against these labels; and (iv) the test sets are about 60 times smaller than VUA, hence the wide intervals. Part of the measured drop is therefore a mismatch in annotation convention, not only model failure. Even so, the errors show genuine domain-shift problems, such as unfamiliar technical vocabulary and informal idioms.

7 Ethical Considerations

The original work VUAMC is built from British National Corpus texts, which are licensed for research. Redistributing derived data therefore depends on BNC and shared-task terms, so we do not redistribute VUA. MelBERT inherits the biases of RoBERTa's web-scale pre-training data. Its notion of "basic meaning" is a single dictionary-based, British/American-English norm, which can misjudge dialectal or culturally specific usage. The paper reports a mean over five seeds and states its compute (two GPUs). Its released code, however, differs from the paper's description in ways that affect results (§4). The honesty and transparency principles of the *Australian Code for the Responsible Conduct of Research* (National Health and Medical Research Council et al., 2018) require such discrepancies to be reported.

Our new data Under the Code's eight principles the main issues are as follows. *Respect and privacy*: review authors posted publicly but did not consent to research use. We use only sentences, drop user identifiers, filter sentences naming people, and release only 150 sentences with labels. Because the reviews are public and nothing is collected from the authors, the risk is low. Publishing this as a paper would still mean asking the university Human Research Ethics Committee whether a negligible-risk exemption applies under the *National Statement* (National Health and Medical Research Council et al., 2023). *Licensing (accountability)*: the Hugging Face copy is labelled Apache-2.0, but the reviews originate from Amazon, whose terms limit reuse. arXiv metadata is CC0. We redistribute only a small research sample with full provenance (`source_doc_id`) and document it with a datasheet-style README (Gebru et al., 2021). *Transparency about annotation*: the second rater was an LLM, not a person. We say so, name the model, report the discarded pilot pass and that adjudication mostly followed the LLM, and release all label sets, so readers can judge the gold standard themselves. Agreement with an LLM is a weaker reliability check than agreement between trained humans, and we treat it as such. *Rigour and honesty*: we report agreement, confidence intervals and negative results. We replaced the synthetic pilot data rather than presenting it as real. *Third-party LLMs*: sending data to an external API discloses it to a provider. This is acceptable only because the text is already public and filtered. LLM labels became gold only after review by the author. *Environmental cost*: training a transformer uses energy (Strubell et al., 2019). We first scored the released checkpoint and limited retraining to a single run. *Open science*: code, labels, guidelines and annotation tools are public, so the work can be checked and extended. *Use of generative AI*: as permitted for this assessment, AI assistance was used for code scaffolding, drafting and editing, as the second annotation rater, and to suggest analyses during adjudication. All experiments were run by the author, and every final label was recorded by the author.

8 Conclusions and Reflections

Replication With three lines of compatibility changes, the official code runs on current software. Retraining gives 78.2 F1 on VUA-18 versus the reported 78.5, so the headline result reproduces to within 0.3 points. The released checkpoint scores 78.8, within 0.3 of the paper. The remaining differences come from seeds and from undoc-

umented choices in the code (test-set evaluation each epoch, a metaphor-class weight of 3) rather than from errors in the method. **Generalisation** On real, MIPVU-annotated product reviews and arXiv abstracts, MelBERT drops to 36.8 and 31.6 F1 (from 72.1 on comparable VUA targets). The main cause is over-prediction on delexical verbs and abstract academic vocabulary, which partly reflects VUA's broad annotation conventions and partly genuine domain shift. MelBERT therefore generalises poorly to these registers when judged against an independent annotation.

Reflections (1) Annotation was the hardest and slowest part. Our pilot pass showed how easy it is to miss conventional metaphors when applying MIPVU, and deciding the "basic" sense of technical vocabulary (*network*, *framework*) needed the most adjudication. A calibration round on VUA examples before annotating, and a second trained human annotator, would strengthen the gold standard. (2) Sampling by sentence and labelling only content words made annotation feasible, but lowered the metaphor rate and ruled out comparison on prepositions, VUA's most metaphor-dense POS. (3) The early synthetic pilot showed the pipeline worked but said nothing about generalisation. Annotating real text from the start would have been better. (4) LLMs were a useful second opinion: LLM1 consistently spotted conventional metaphors the human missed. But two LLMs given identical instructions found 8.0% and 1.3% of targets metaphorical, and changed MelBERT's estimated F1 from 32 to 7. They should guide human review, not replace it. (5) Time pressure limited hand annotation to 40 of the 150 sentences. Annotating the rest, and calibrating against VUA examples, would narrow the confidence intervals and separate convention mismatch from true domain shift.

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